

Quiz (Carolina Reaper)

- Q1.** A logistics drone is programmed to maintain a distance d from a linear pipeline such that $|y - 12| = 5$. If y represents the drone's lateral coordinate in meters, what are the possible coordinates?
- (A) $\{7, 17\}$
(B) $\{-7, 17\}$
(C) $\{5, 12\}$
(D) $\{-17, -7\}$
(E) $\{12, 17\}$
- Q2.** The deviation of a cargo ship's fuel consumption f from the optimal rate R is expressed as $|f - R|$. If $R = 500$ and $|f - 500| \leq 25$, which interval represents the acceptable fuel consumption?
- (A) $[475, 500]$
(B) $[500, 525]$
(C) $[475, 525]$
(D) $\{475, 525\}$
(E) $(-\infty, 525]$
- Q3.** Two delivery trucks T_1 and T_2 are located at mile markers m_1 and m_2 on a straight highway. If $|m_1 - m_2| = |m_1| + |m_2|$ and $m_1, m_2 \neq 0$, which statement regarding their positions relative to the dispatch center at 0 is true?
- (A) They are on the same side of the center.
(B) They are at the same location.
(C) They are on opposite sides of the center.
(D) One truck is at the dispatch center.
(E) The distance between them is zero.
- Q4.** A sorting facility uses the expression $E = ||x - 10| - |x - 20||$ to calculate throughput error. If a package is at position $x = 15$, evaluate E .
- (A) 10
(B) 5
(C) 0
(D) -5
- (E) 15
- Q5.** In a traffic flow model, the velocity v of vehicles must satisfy $|2v - 110| < 20$. What is the maximum integer velocity allowed?
- (A) 65
(B) 64
(C) 45
(D) 46
(E) 55
- Q6.** A temperature-controlled container must stay within a range defined by $|T - C| = k$. If the maximum temp is 8°C and the minimum is -2°C , find the values of C and k .
- (A) $C = 5, k = 3$
(B) $C = 3, k = 5$
(C) $C = 6, k = 2$
(D) $C = 2, k = 6$
(E) $C = 0, k = 8$
- Q7.** A courier travels from A to B . The distance is given by $d(x, y) = |x - y|$. If A is at -45 and the courier travels to B such that $d(A, B) = 100$, what is the sum of all possible coordinates for B ?
- (A) 100
(B) 55
(C) -90
(D) -45
(E) 145
- Q8.** Simplify the cost function $C(x) = |x - 5| + |x - 10|$ for a warehouse located at x where $5 < x < 10$.
- (A) $2x - 15$
(B) $15 - 2x$
(C) 5
(D) -5
(E) $2x + 5$

Open Ended Questions (FRQ)

- Q9.** A delivery van starts at the distribution hub (coordinate 0), drives to a drop-off point at 18 km, then to a pick-up point at -7 km, and finally returns to the hub. Write an expression using absolute values to represent the total distance traveled and calculate the final result.
- Q10.** A shipping schedule is considered 'on-time' if $|t_{\text{actual}} - t_{\text{scheduled}}| \leq 0.5$ hours. If a truck arrives at 14.2 hours for a 15.0 hour slot, determine if it is on-time by evaluating the absolute value expression. Show your steps.

Chef's Tips

- Emphasize that absolute value represents undirected distance on the number line.
- Challenge students to interpret $|a - b|$ as the distance between a and b before using algebra.
- Discuss the physical implications of $|x + y| \leq |x| + |y|$ in logistics displacement versus total distance.